Mess Automation System

Seamless dining management with automated billing and face recognition

Github Link

Presented by : Team Byte Busters

Need for the Software

- **The Queue Bottleneck:**
  Long wait times and crowding at physical registers during peak meal hours.

- **The Rebate Runaround:**
  Tedious, paper-based application processes for mess leaves, with zero visibility on approval status.

- **Billing "Blind Spots":**
  Inability to track daily extra purchases, leading to end-of-month "sticker shock" and billing disputes.

Need for the Software

- **Vulnerable Authentication:**
  Relying on visual checks makes unauthorised access.

- **The Accounting Nightmare:**
  Staff waste days manually tallying hundreds of messy, handwritten register entries to calculate monthly bills.

Potential Users

- Students / Hostel Residents
- Mess Managers

MAJOR REQUIREMENTS

Functional Requirements

- **Biometric Authentication:** Integration of a face recognition system for fast, touchless student login and access control.
- **Automated Billing & Tracking:** Real-time logging of daily meals consumed and automated generation of accurate student bills.
- **Role-Based Access:** Distinct secure interfaces and permissions for Administrators (managing operations) and Students (viewing usage and dues).
- **Centralized Database Management:** Ability for admins to seamlessly add/remove users, update pricing, and modify mess records.

Non-Functional Requirements

- **Performance:** High-speed processing for facial recognition and database queries to eliminate queues during peak dining hours.
- **Security:** Secure handling of sensitive biometric data and robust protection of financial records.
- **Reliability:** Consistent uptime and data integrity, ensuring no meals go unlogged and no billing errors occur.

Implementation

Frontend :

- Built with React.js and TypeScript to provide role-based, dynamic dashboards and real-time webcam integration for facial capture.

Backend :

- Developed using Node.js and Express.js following an MVC architecture to handle RESTful APIs and process secure authentication.

Database :

- Utilizes PostgreSQL for ACID-compliant storage of student records, billing data, and biometric snapshots stored as JSON/Base64 strings.

Authentication :

- Features secure login via institute credentials and a custom Face Recognition system with role-based access control.

Design

Model View Architecture

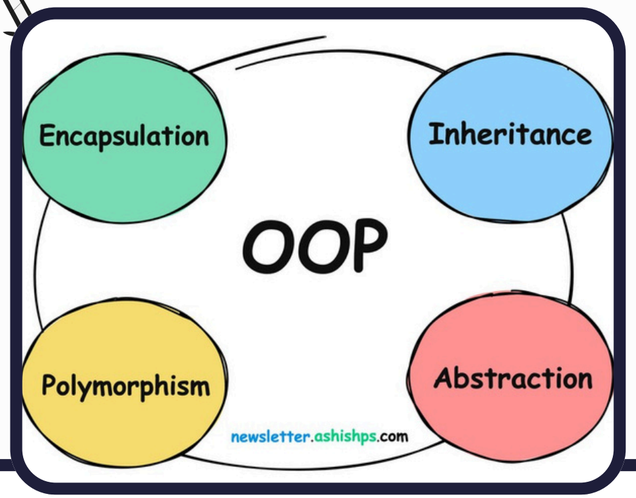
Multiple views for common data access

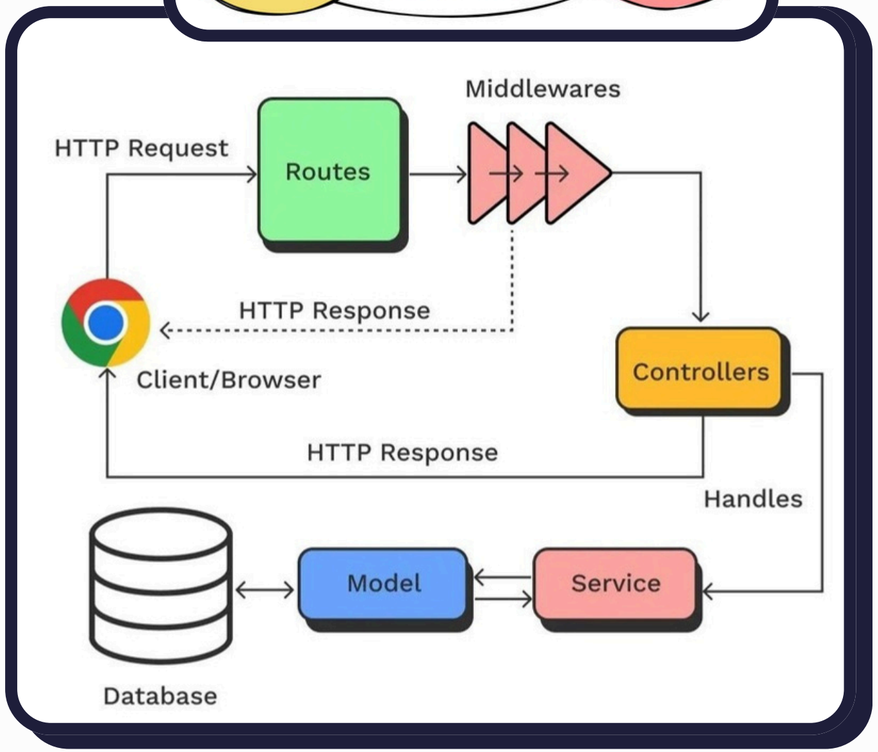
- **Object Oriented Design**

Encapsulated, modular, and reusable classes

- **Page-wise Views and Controllers**

Separate logic for each feature page





DEMONSTRATION

WEBSITE

http://172.27.16.252:5176/

LINK

Future Development Plans

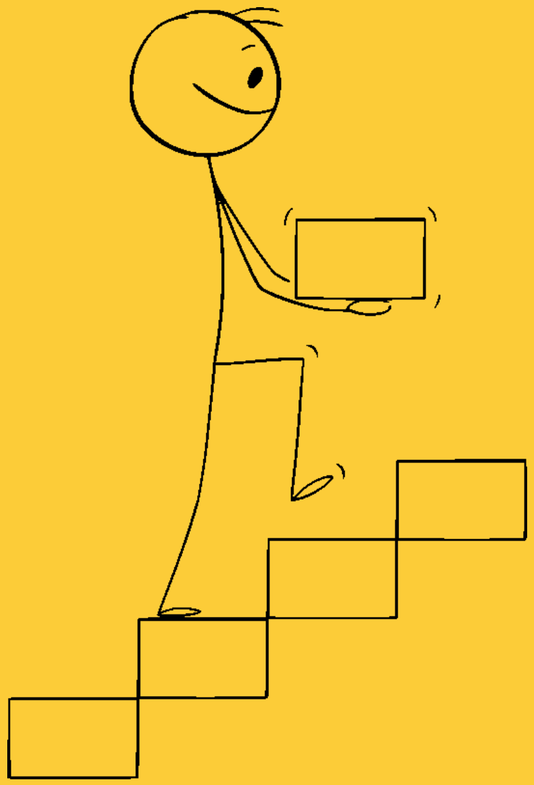
SCALABILITY
Extend the system to support multiple hostels and institutions with a centralized architecture.

MOBILE APPLICATION
Develop Android/iOS apps for easier access, real-time updates, and improved user experience.

AI-BASED ANALYTICS
Use data-driven insights to predict food demand, reduce wastage, and optimize menu planning.

ONLINEPAYMENTS
Integrate UPI and other payment methods for seamless mess bill transactions.

ENHANCED SECURITY
Improve face recognition accuracy and implement stronger authentication and data protection.



Learnings...

- Clarity before execution
- Working effectively in a team
- Adapting to challenges
- Managing time efficiently

TEAM

B Mahath	240252
Priyanshi Meena	240804
Rishith Jalagam	240484
Eranki Venkata Jhanani	240385
Basavakiran Shivanand Deshmukh	240261
Inapakurti Rajesh	240463
Parth Parashar	240735
Shreyansh Dewangan	230987
Rohan Kumar	230865
Siddhant Singh	231004

ACKNOWLEDGEMENT

We would like to express our deepest gratitude to everyone who supported us throughout the development of the Mess Automation System. This project has been a significant learning experience, allowing us to bridge the gap between theoretical computer science and practical, real-world application.

We wish to extend our sincere thanks to Professor Indranil Saha for the invaluable guidance, encouragement, and the opportunity to work on this project. The insights into software engineering principles and system design were instrumental in helping us structure our application effectively and overcome complex technical challenges.

We are also grateful to our Project Mentor for their continuous support and technical feedback. Their timely assistance during lab sessions and their clarity in addressing our queries regarding database management and system integration were vital to the successful completion of this project.

THANK YOU